RAG-Based Customer Support Assistant (with LangGraph + HITL)

TECHNICAL DOCUMENTATION

🟢 1. INTRODUCTION

🔹 What is RAG?

Retrieval-Augmented Generation (RAG) is a technique that combines:

- Information retrieval from a knowledge base
- Text generation using a model

It improves accuracy by using real data instead of relying only on pre-trained knowledge.

🔹 Why RAG is Needed

- Traditional chatbots give generic answers
- They may produce incorrect information
- RAG provides context-based accurate answers

🔹 Use Case

This project implements a Customer Support Assistant that:

- Reads information from a PDF
- Answers user queries
- Escalates complex queries to human support

🟢 2. SYSTEM ARCHITECTURE EXPLANATION

The system consists of the following components:

- PDF Loader → loads document
- Chunking → splits text
- Embedding Model → converts text into vectors
- Vector Database (ChromaDB) → stores embeddings
- Retriever → fetches relevant data
- Answer Generator → generates response
- LangGraph → manages workflow
- HITL → human escalation

◆ Working

1. PDF is processed
2. Data is stored as embeddings
3. User sends query
4. Relevant data is retrieved
5. Answer is generated
6. If needed → escalated to human

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3. DESIGN DECISIONS

◆ Chunk Size

- Used small chunks (~500 characters)
- Ensures better retrieval accuracy

◆ Embedding Strategy

- Used HuggingFace embeddings (free)
- Avoids dependency on paid APIs

◆ Retrieval Approach

- Similarity-based search using ChromaDB

- Retrieves most relevant chunks

🔹 Prompt Design

- Structured prompts used for answer generation
- Focus on concise responses

🟢 4. WORKFLOW EXPLANATION

The system uses a simple workflow:

Input → Retrieve → Generate → Output

🔹 Steps

1. User enters query
2. System retrieves relevant chunks
3. Answer is generated
4. Output is returned

🟢 5. CONDITIONAL LOGIC

The system uses decision-making:

- If answer found → return response
- If no answer → escalate
- If query unclear → escalate

🟢 6. HITL IMPLEMENTATION

🔹 Role of Human

- Handles complex queries
- Provides accurate responses

🔹 Benefits

- Improves reliability
- Avoids incorrect answers

🔹 Limitations

- Slower response
- Requires human availability

🟢 7. CHALLENGES & TRADE-OFFS

🔹 Retrieval Accuracy vs Speed

- More chunks → better accuracy but slower

🔹 Chunk Size vs Context

- Small chunks → precise retrieval
- Large chunks → better context

🔹 Cost vs Performance

- Free models → lower quality
- Paid models → better accuracy

🟢 8. TESTING STRATEGY

🔹 Approach

- Tested with multiple queries

🔹 Sample Queries

- What is refund policy?
- How to contact support?
- What are working hours?

- Result
  - System successfully retrieves and answers queries

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9. FUTURE ENHANCEMENTS

- Support multiple PDFs
- Add memory for conversation
- Improve answer generation
- Deploy as web application